Aden Walsey

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EDUCATION

Cornell University, College of Engineering, Ithaca, NY
Master of Engineering in Mechanical Engineering, Early Admit, Arps Fellow, **GPA: 4.0** **Expected December 2026**
Bachelor of Science in Mechanical Engineering, Dean's List, Varsity Rowing **May 2026**

SKILLS

CAD & Simulation: SolidWorks, Autodesk Fusion 360, ANSYS (FEA/CFD).
Analysis & Programming: GD&T, Tolerance Analysis, MATLAB/Simulink, Python.
Manufacturing & Testing: CNC Machining, DFM, Rapid Prototyping, Technical Drawings, Qualification Testing.
Leadership: Cornell Varsity Rowing Coxswain: Second Team All-Ivy (2025), Head of the Charles 1st Place 1V4+ (2023).

PROJECTS

Spacecraft Thruster Test Assembly Redesign, *Advisor Stefan Bell,* Cornell ASTRA Lab **August 2026 – Present**

- Redesigning, fabricating, and testing an electrospray thruster test assembly for spacecraft propulsion research.
- Refining component alignment and assembly reliability to reduce propellant-induced electrical shorts during testing and increase test repeatability**.**

Bus Shelter Base Design, *Advisor Sirietta Simoncini*, CUSD Sustainable Mobility Project Team **Spring 2026**

- Led the base assembly design of a sustainable, modular bus shelter using SolidWorks, considering material selection, fabrication methods, structural interfaces, and integration with the overall shelter system.
- Performed single-module FEA in ANSYS using New York State building code loading requirements for dead, wind, and snow conditions, incorporating material properties and weldment connections to evaluate performance.

Bicycle Frame Structural Design & Optimization, *Advisor Rajesh Bhaskaran,* Cornell University **Spring 2026**

- Redesigned a conventional aluminum bicycle frame into a step-through e-bike in Fusion 360, balancing structural requirements, accessibility, battery integration, and fabrication constraints.
- Used ANSYS FEA to iterate tube geometry and wall thickness after removing the top tube and implementing battery loading, maintaining peak stress within 1% of the baseline design.

PROFESSIONAL EXPERIENCE

Mechanical Engineering Intern, *Princeton CarbonWorks.* Mystic, CT **Summer 2026 – Present**

- Developed high-performance cycling hardware through iterative CAD, simulation, prototyping, and physical testing.
- Collaborating with Trek Bicycles' Lidl-Trek racing team to design a modular aerobar positioning system in SolidWorks for WorldTour time trial bicycles, replacing custom 3D-printed parts with standardized CNC components.
- Built a parametric bicycle wheel model in SolidWorks and performed CFD in NablaFlow to optimize performance across yaw angles weighted for the Ironman World Championship course, reducing drag from baseline by 36%.

Manufacturing Engineering Intern, *Mainspring Energy.* Menlo Park, CA **Summer 2025**

- Designed production tooling and developed hardware qualification tests for linear-generator manufacturing.
- Built and validated a pneumatic stator tooling fixture using SolidWorks to pierce insulation without damaging copper, using calculated engagement depth, tolerances, and replaceable components to reduce process time by 60%.
- Planned and executed instrumented drop tests on a reusable stator shipping crate, identified loads above validated baselines, and designed a revised installation method to prevent stator-to-stator collisions.

R&D Engineering Intern, *Neodyne Biosciences, Inc.* Fremont, CA **Summer 2024**

- Supported medical device manufacturing improvements by developing adhesion test methods to investigate assembly failures, reducing production defects by over 70%.
- Performed tolerance analysis on device packaging to improve manufacturability and streamline assembly, creating SolidWorks drawings to communicate design improvements to manufacturing.

INTERESTS

Cycling, Backpacking, Cooking, Motorsport, Drawing, Hi-Fi Audio Systems, Furniture Design, Film Photography.